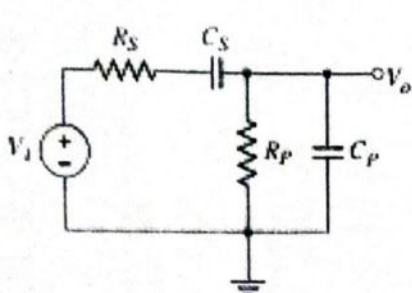
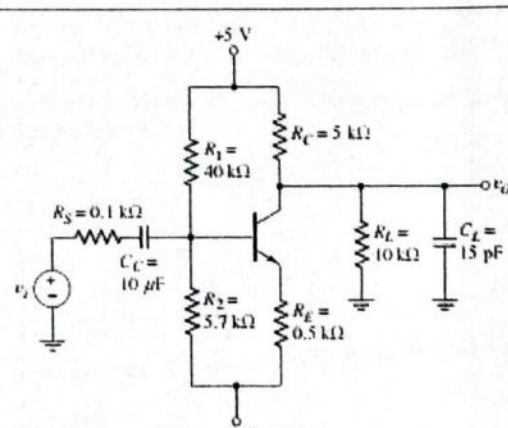
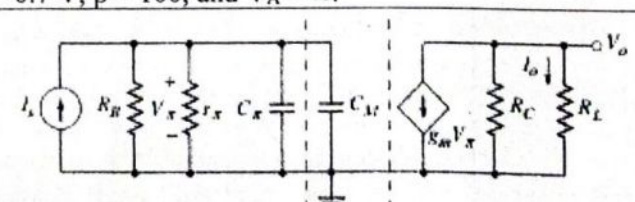




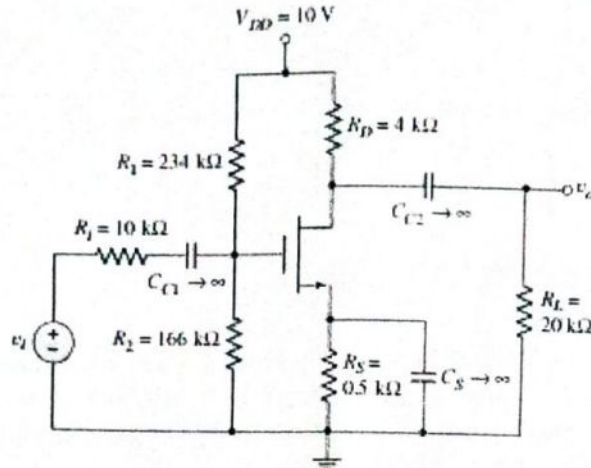
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded.
Department of Electronics and Telecommunication
Academic Year 2024-25 Semester 1 Class: SY (EXTC) ISE-2

Subject: - PCC-EC-201 Electronic Devices and Circuit Duration: 45 minutes Max. Marks: 20
Date: 14/10/2025 Time: 14:30 AM

Note: - Assume suitable data wherever necessary. The last column indicates marks. Use of nonprogrammable calculators allowed.

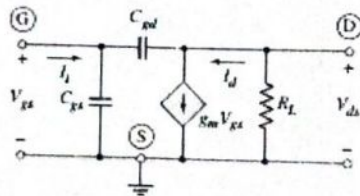
Q. No.	Question	BL	M
Q1A	 The value of R_S in the circuit in Figure is $R_S = 1 \text{ k}\Omega$. The midband gain is -1 dB, and the corner frequencies are $f_L = 100 \text{ Hz}$ and $f_H = 1 \text{ MHz}$ (a) Determine R_P, C_S, and C_P. (b) Determine the open-circuit and short-circuit time constants.	3	5
Q1B	 Determine the midband gain, corner frequencies, and bandwidth of a circuit containing both a coupling capacitor and a load capacitor. Consider the circuit shown in Figure with transistor parameters $V_{BE(on)} = 0.7 \text{ V}$, $\beta = 100$, and $V_A = \infty$.	4	5
Q2A	 Determine the 3 dB frequency of the current gain for the circuit shown in Figure, both with and without the effect of C_M. The circuit parameters are $R_C = R_L = 4 \text{ k}\Omega$, $r_\pi = 2.6 \text{ k}\Omega$, $R_B = 200 \text{ k}\Omega$, $C_\pi = 4 \text{ pF}$, $C_\mu = 0.2 \text{ pF}$, and $g_m = 38.5 \text{ mA/V}$.	3	5

Q2B

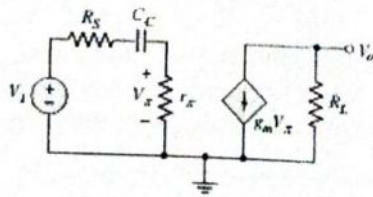


For the circuit in Figure, the transistor parameters are: $K_n = 0.5 \text{ mA/V}^2$, $V_{TN} = 2 \text{ V}$, $\lambda = 0$, $C_{gd} = 0.1 \text{ pF}$, and $C_{gs} = 1 \text{ pF}$. Calculate: (a) the Miller capacitance, and (b) the 3 dB frequency of the small-signal voltage gain.

2(Bonus question)



Derive expression for Miller capacitance, and f_T .



For the equivalent circuit shown in Figure, the parameters are: $R_S = 1 \text{ k}\Omega$, $r_x = 2 \text{ k}\Omega$, $R_L = 4 \text{ k}\Omega$, $g_m = 50 \text{ mA/V}$, and $C_C = 1 \text{ }\mu\text{F}$. (a) Determine the expression for the circuit time constant. (b) Calculate the 3 dB frequency and maximum gain asymptote. (c) Sketch the Bode plot of the transfer function magnitude.

S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
ISE-II Examination AY 2025-26

Class : SY

Date: 16/10/2025 Time: 14:30-15:15 Exam duration : 45 min	Subject: Fundamentals of Digital Image processing Course Code: MDM-IS-EC201	Max Marks: 20
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Note:

- 1) All questions are compulsory.
- 2) Assume suitable data if necessary.
- 3) Use of calculator is permitted.

Q.No	Questions	Marks	COs	BT
Q1	Compute the DCT for the given 4x4 image $[A] = \begin{bmatrix} 2 & 1 & 1 & 2 \\ 2 & 2 & 1 & 1 \\ 1 & 1 & 2 & 2 \end{bmatrix}$	7	CO1	BT1
Q2	Define Histogram and Perform Histogram Equalization for the following 2 bit image. $I(x,y) = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 2 & 3 & 3 & 1 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 \end{bmatrix}$	6	CO3	BT1, BT3
Q3	What is the value of rounded pixel if it is smoothened by a 3x3 box filter and weighted average filter? $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	7	CO3, CO5	BT2, BT3

Blooms' Taxonomy:

BT1: REMEMBER, BT2: UNDERSTAND, BT3: APPLY, BT4: ANALYSIS, BT5: EVALUATE, BT6: CREATE

Course Outcome (Cos):

CO1. Review the fundamental concepts of a digital image processing system and representation techniques

CO2. Analyse Images in the frequency domain using Fourier transforms.

CO3. Evaluate the techniques for image enhancement.

CO4. Categorize various enhancement techniques and interpret Image compression standards.

CO5. Evaluate the techniques for Image restoration standards.



Shri Guru Gobind Singh Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Electronics and telecommunication
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: SY B. Tech

Course: Network Theory

Course Code: PCC-EC203

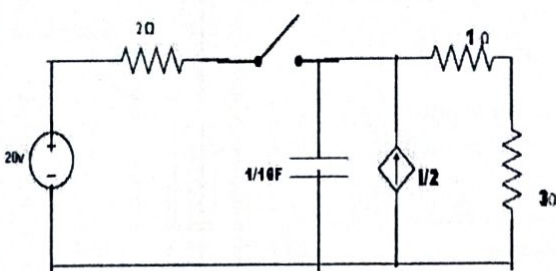
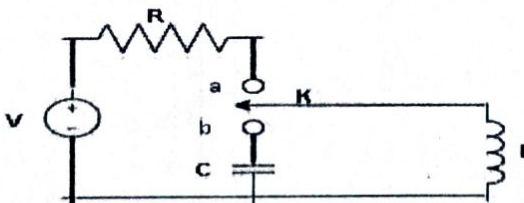
Date: 13/10/2025

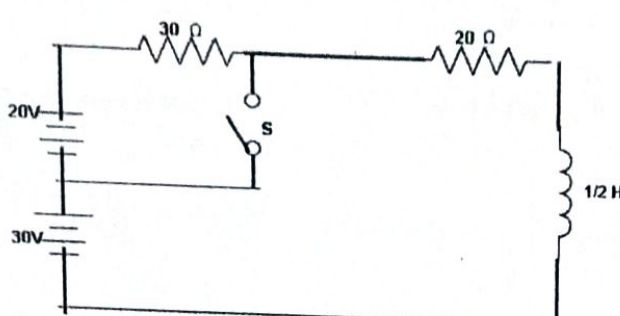
Time: 2:30 pm to 03:15pm

Max. Marks: 20

Note:

- Mobile phones and programmable calculators are strictly prohibited.
- Figures to right indicate full marks.

Q.N.1	SOLVE ALL THE QUESTIONS.	Marks	CO	BT
a.	Explain initial conditions in elements R, L, C with equivalent circuit at $t = 0$ and $t = \infty$ in transient analysis.	2	CO3	BT2
b.	Explain the term driving force and driving point with neat diagram.	2	CO4	BT2
c.	Draw and explain frequency domain equivalent circuits for resistor, inductor, and capacitor.	2	CO5	BT1
Q.N.2	SOLVE ALL THE QUESTIONS.			
a.	Determine complete expression for current i with respect t using transient analysis in time domain ($R_1=2\Omega, R_2=1\Omega, R_3=3\Omega, C=1/16F, V=20V$, dependent current $I/2$) 	7	CO3	BT5
b.	In the circuit shown switch S moves from terminal a to terminal b. At $t=0$ find the expression of $I(t)$ using Laplace transform. 	7	CO3	BT5

OR			
c.	In the circuit shown switch S is OFF for long time. At $t=0$, switch S is ON. find the expression of $I(t)$ using Laplace transform. ($R_1=30\Omega$, $R_2=20\Omega$, $L=1/2H$, $V_1=20V$, $V_2=30V$) 	7	CO3 BT5

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand basics electrical circuits with nodal and mesh analysis.
CO 2	Appreciate electrical network theorems.
CO 3	Apply Laplace Transform for steady state and transient analysis.
CO 4	Determine different network functions.
CO 5	Appreciate the frequency domain techniques.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Instrumentation Engineering

In-Semester Examination II

Academic Year: 2025-26 (Semester I)

Class: SY B.Tech.

Course: Electronic Devices and Circuits

Course Code: PCC-IN202

Date: 11.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve any Four of the following.

Q.N.		Marks	CO	BT
Q.1	Give detailed classification of multistage amplifier.	05	CO6	BT1
Q.2	With the help of neat circuit diagram explain RC coupled amplifier.	05	CO6	BT1
Q.3	For RC coupled amplifier, $V_{CC} = 12V$, $R_1 = 100\text{ k}\Omega$, $R_2 = 22\text{ k}\Omega$, $R_C = 4.7\text{ k}\Omega$, $R_E = 1\text{ k}\Omega$, $V_{BE} = 0.7$, $\beta = 100$, Input coupling capacitor $C_i = 10\mu\text{F}$ and output coupling capacitor $C_o = 10\mu\text{F}$, $R_L = 10\text{ k}\Omega$ (connected after output coupling capacitor), $r_e = 17.8\text{ }\Omega$. Perform its DC analysis and also find out the voltage gain A_v .	05	CO3	BT5
Q.4	What is power amplifier? List out various class of power amplifier. Also explain characteristics of class A amplifier.	05	CO6	BT2
Q.5	Discuss Push-Pull Class A Power Amplifier.	05	CO6	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To remember the basic electronic components and observe various characteristics of the same.
CO 2	To understand the various theorems, utilizing equivalent circuits or developing models and applying the fundamental circuit theorems rather than memorizing the equations.
CO 3	To practice different biasing circuits using equivalent models to illustrate various circuit parameters.
CO 4	To analyze and design basic electronic circuits, particularly with application to diodes, MOS field-effect transistors, bipolar junction transistors.
CO 5	To measure the characteristics of two port network.
CO 6	To design and construct different electronic circuitry like single-stage amplifier, multi-stage amplifier, oscillator circuits, multivibrators, feedback amplifiers.